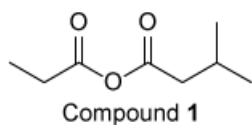


Compound 1 is shown below.



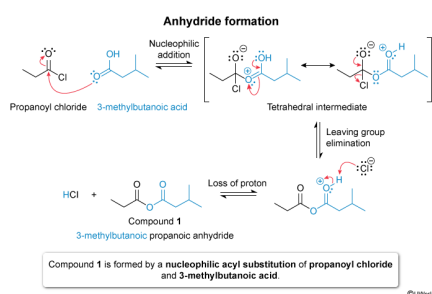
Compound 1 is prepared by:

- ☐ A. condensation of propanoic acid with butanoic acid. [14%]
- ☐ B. nucleophilic acyl substitution of 3-methylbutanoic propanoic anhydride with 1-propanol. [10%]
- ☐ C. nucleophilic acyl substitution of propanoyl chloride and 3-methylbutanol. [8%]
- ✔ ☒ D. nucleophilic acyl substitution of propanoyl chloride and 3-methylbutanoic acid. [66%]

Correct

65% answered correctly

Explanation:



Carboxylic acid anhydrides are **carboxylic acid derivatives** that result from the **condensation** of two carboxylic acids, resulting in the loss of H_2O (ie, dehydration). In addition, acid anhydrides can be synthesized by **nucleophilic acyl substitution** between an **acid halide** (the most reactive carboxylic acid derivative) and a **carboxylic acid**. The carboxylic acid acts as the nucleophile and adds to the acid chloride carbonyl, creating a tetrahedral intermediate. After the chloride leaving group is eliminated, a proton is lost, yielding an anhydride.

Compound 1 is an asymmetrical (mixed) anhydride because it is formed from two different carboxylic acids (or a carboxylic acid and acid chloride containing different alkyl groups). The carboxylic acid and acid chloride used to prepare an anhydride must contain the correct number of carbons and substituents on each half of the anhydride. One half of Compound 1 contains a three-carbon chain, and the other half contains a four-carbon chain with a methyl substituent on carbon 3. Therefore, Compound 1 could be synthesized by either:

- nucleophilic acyl substitution of propanoyl chloride and 3-methylbutanoic acid
- nucleophilic acyl substitution of propanoic acid and 3-methylbutanoyl chloride
- condensation of propanoic acid and 3-methylbutanoic acid

The only choice that gives the correct reaction and starting material pairing is a **nucleophilic acyl substitution of propanoyl chloride and 3-methylbutanoic acid**.

(Choice A) Although anhydrides can be prepared by the condensation of two carboxylic acids, butanoic acid is missing the methyl substituent on carbon 3.

(Choice B) Nucleophilic acyl substitution of 3-methylbutanoic propanoic anhydride (ie, Compound 1) and 1-propanol yields an **ester**. The question asks how to *prepare* Compound 1 rather than how to use it to make another carboxylic acid derivative.

(Choice C) Although the starting materials propanoyl chloride and 3-methylbutanol contain the same alkyl chains as Compound 1, nucleophilic acyl substitution of an acid chloride and an *alcohol* yields an ester instead of an anhydride.

Educational objective:

Carboxylic acid anhydrides are carboxylic acid derivatives made from the condensation of two carboxylic acids, resulting in the loss of H_2O (dehydration). In addition, anhydrides can be synthesized by nucleophilic acyl substitution of an acid

chloride and a carboxylic acid.

Time spent:0 sec

Subject:
Organic Chemistry

QID:403895

System:
Functional Groups and Their Reactions